
Monte Carlo Simulations of Yang-Mills Matrix Model

Project Report



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Abstract

In this project we have simulated zero-dimensional Yang-Mills matrix model with gauge group $SU(N)$ for $N = 4, 8, 9, 10, 12$, using the method of Markov Chain Monte Carlo. We calculated two observables of the model, the expectation value of the action and the extent of spacetime. Our simulation results are in excellent agreement with the analytical results.

Acknowledgements

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Last but not the least, I would like to expand my thanks to everyone who directly or indirectly helped me in completing this project.

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Contents

1	Yang-Mills Matrix Model	1
1.1	Yang-Mills action and dimension reduction	1
1.2	Chern-Simons action	3
1.3	Theoretical results	4
2	Markov Chain Monte Carlo	5
2.1	Importance sampling	5
2.2	Metropolis algorithm	5
2.3	Statistical errors	7
3	Simulation results	8
3.1	Average action $\langle S \rangle$	8
3.2	Extent of spacetime R	11
4	Discussions and conclusions	16
	References	17

List of Figures

3.1	Acceptance rate against Monte Carlo time.	8
3.2	Plots showing thermalization of S with Monte Carlo time . .	9
3.3	Plot of $\frac{\langle S \rangle}{N^2 - 1}$ against d for different values of N	10
3.4	Combined plot of $\frac{\langle S \rangle}{N^2 - 1}$ against d for different values of N . .	11
3.5	Plots showing thermalization of $\text{Tr}(X_i^2)$ with Monte Carlo time.	12
3.6	Plot of $\frac{\langle \text{Tr}(X_i^2) \rangle}{N}$ against d for different values of N	13
3.7	Combined plot of $\frac{\langle \text{Tr}(X_i^2) \rangle}{N}$ against d for different values of N	14
3.8	The extent of spacetime R against N for various values of d . The solid lines show linear fits to the data. We see that the slope $\omega \lesssim \frac{1}{4}$, as predicted by the behavior in Ref. [3] . .	15

1 Yang-Mills Matrix Model

1.1 Yang-Mills action and dimension reduction

In four-dimensional Minkowski spacetime with metric $g_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$, Yang-Mills action with Chern-Simons term is given by

$$S = - \int d^4x \left(\frac{1}{2g^2} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) + \frac{\theta}{16\pi^2} \text{Tr}(F_{\mu\nu} \tilde{F}^{\mu\nu}) \right), \quad (1.1)$$

where we define

$$D_\mu = \partial_\mu - i[A_\mu, \cdot], \quad (1.2)$$

$$F_{\mu\nu} = \partial_{[\mu} A_{\nu]} - i[A_\mu, A_\nu], \quad (1.3)$$

$$\tilde{F}^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\alpha\beta} F_{\alpha\beta}. \quad (1.4)$$

The path integral of interest is

$$Z = \int [\mathcal{D}A_\mu] \exp(iS). \quad (1.5)$$

This is invariant under the finite gauge transformations

$$A_\mu \longrightarrow g^{-1} A_\mu g + i g^{-1} \partial_\mu g, \quad (1.6)$$

with $g \in SU(N)$.

Now we Wick rotate to Euclidean signature i.e.,

$$x^0 \longrightarrow x^4 = i x^0 \quad (1.7)$$

and as a consequence

$$d^4x \longrightarrow d_E^4x = i d^4x, \quad (1.8)$$

$$\partial_0 \longrightarrow \partial_4 = -i \partial_0, \quad (1.9)$$

and because now the metric become $g_{\mu\nu} = \text{diag}(-1, -1, -1, -1)$ so for the correct inner product,

$$A_0 \longrightarrow A_4 = -i A_0. \quad (1.10)$$

From here we conclude that

$$F_{\mu\nu}F^{\mu\nu} \longrightarrow (F_{\mu\nu}^2)_E. \quad (1.11)$$

and

$$F_{\mu\nu}\tilde{F}^{\mu\nu} \longrightarrow i(F_{\mu\nu}\tilde{F}^{\mu\nu})_E. \quad (1.12)$$

So the action and path integral becomes

$$Z_E = \int [\mathcal{D}A_\mu] \exp(-S_E), \quad (1.13)$$

$$S_E = \int d^4x_E \left(\frac{1}{2g^2} \text{Tr}(F_{\mu\nu}^2)_E + \frac{i\theta}{16\pi^2} \text{Tr}(F_{\mu\nu}\tilde{F}^{\mu\nu})_E \right). \quad (1.14)$$

From above it is clear that the Chern-Simons term is purely imaginary. From now on we will drop all the E subscripts. Now let us first consider the pure Yang-Mills term.

$$S_{YM} = \frac{1}{2g^2} \int d^4x \text{Tr}(F_{\mu\nu}^2). \quad (1.15)$$

From the variation of the above action we get

$$\delta S_{YM} = -\frac{2}{g^2} \int d^4x \text{Tr}(D_\mu F_{\mu\nu} \delta A_\nu) + \frac{2}{g^2} \int d^4x \text{Tr}(\partial_\mu (F_{\mu\nu} \delta A_\nu)). \quad (1.16)$$

So for the field whose variation vanishes at infinity, the equation of motion is given by

$$D_\mu F_{\mu\nu} = 0 \quad \text{equivalently} \quad \partial_\mu F_{\mu\nu} - i[A_\mu, F_{\mu\nu}] = 0. \quad (1.17)$$

We can reduce to zero dimensions by upgrading vector fields to scalar fields and these new scalar fields are independent to spacetime coordinates. So $A_\mu \longrightarrow X_\mu$ and action and equation of motion becomes

$$S_{YM} = -\frac{V_{\mathbb{R}^4}}{2g^2} \text{Tr}([X_\mu, X_\nu]^2), \quad (1.18)$$

$$[X_\mu, [X_\mu, X_\nu]] = 0. \quad (1.19)$$

1.2 Chern-Simons action

Now let us consider the Chern-Simons term, the corresponding Lagrangian after suppressing the $i\theta$ is

$$\mathcal{L}_\theta = \frac{1}{16\pi^2} \text{Tr}(F_{\mu\nu} \tilde{F}^{\mu\nu}). \quad (1.20)$$

The variation in the Lagrangian is given by

$$\delta\mathcal{L}_\theta = \frac{1}{8\pi^2} \epsilon_{\mu\nu\alpha\beta} \text{Tr}(F_{\mu\nu} D_\alpha(\delta A_\beta)). \quad (1.21)$$

Using the Jacobi identity $\epsilon_{\beta\alpha\mu\nu} D_\alpha F_{\mu\nu} = 0$, the above variation becomes

$$\delta\mathcal{L}_\theta = \frac{1}{8\pi^2} \epsilon_{\mu\nu\alpha\beta} \partial_\alpha \text{Tr}(F_{\mu\nu} \delta A_\beta) = \partial_\alpha \delta\mathcal{K}_\alpha, \quad (1.22)$$

where $\delta\mathcal{K}_\alpha = \frac{1}{8\pi^2} \epsilon_{\mu\nu\alpha\beta} \text{Tr}(F_{\mu\nu} \delta A_\beta)$. This clearly shows that Chern-Simons term does not contribute to the equation of motion. Using this we can find the current $\mathcal{K}_\alpha$ and it turns out to be

$$\mathcal{K}_\alpha = \frac{1}{16\pi^2} \epsilon_{\alpha\mu\nu\beta} \text{Tr} \left(F_{\mu\nu} A_\beta + i \frac{2}{3} A_\mu A_\nu A_\beta \right). \quad (1.23)$$

Now this theta term (Chern-Simons term) is proportional to an integer k known as *topological charge*. It is defined by

$$k = \int d^4x \mathcal{L}_\theta = \int d^4x \partial_\alpha \mathcal{K}_\alpha. \quad (1.24)$$

So the Chern-Simons action becomes

$$S_{CS} = i\theta k. \quad (1.25)$$

Now $\partial\mathbb{R}^4 = S_\infty^3$, and using the generalized divergence theorem we get

$$k = \int_{\partial\mathbb{R}^4=S_\infty^3} d^3\sigma_\alpha \mathcal{K}_\alpha. \quad (1.26)$$

For finite action $F_{\mu\nu}(x) = o(1/x^2)$ as $x \rightarrow \infty$, then only second term contribute to the action

$$k = \frac{i}{24\pi^2} \epsilon_{\alpha\mu\nu\beta} \int_{\partial\mathbb{R}^4=S_\infty^3} d^3\sigma_\alpha \text{Tr}(A_\mu A_\nu A_\beta). \quad (1.27)$$

By using the fact that $A_\mu = iX_\mu$ on S_∞^3 we have

$$k = \frac{1}{24\pi^2} \epsilon_{\alpha\mu\nu\beta} \int_{\partial\mathbb{R}^4=S_\infty^3} d^3\sigma_\alpha \text{Tr}(X_\mu X_\nu X_\beta). \quad (1.28)$$

Solving the above integral by local parameterization $\xi_a = \xi_a(x)$ of the group G with $X_a = g^{-1}\partial_a g$, we get

$$k = \frac{1}{24\pi^2} \int_{\partial\mathbb{R}^4=S_\infty^3} d^3\xi \epsilon_{abc} \text{Tr}(X_a X_b X_c). \quad (1.29)$$

By reducing the dimension to zero, we get

$$S_{CS} = \frac{i\theta V_{S^3}}{24\pi^2} \epsilon_{abc} \text{Tr}(X_a X_b X_c) \quad a, b, c \text{ runs from } 1, \dots, 3. \quad (1.30)$$

By combining both the terms we get

$$S_E = -\frac{V_{\mathbb{R}^4}}{2g^2} \text{Tr}([X_\mu, X_\nu]^2) + \frac{i\theta V_{S^3}}{24\pi^2} \epsilon_{abc} \text{Tr}(X_a X_b X_c). \quad (1.31)$$

We can do the rescaling and can write $\beta = \frac{V_{\mathbb{R}^4}}{2g^2}$ and $\alpha = \frac{\theta V_{S^3}}{24\pi^2}$ and the action becomes

$$S = -\beta \text{Tr}([X_\mu, X_\nu]^2) + i\alpha \epsilon_{abc} \text{Tr}(X_a X_b X_c). \quad (1.32)$$

Important: We can only handle real actions from Metropolis Monte Carlo simulation so only pure Yang-Mills term will be considered during simulation.

1.3 Theoretical results

This exact result holds when we consider the pure Yang-Mills term only

$$\frac{\langle S \rangle}{N^2 - 1} = \frac{d}{4}, \quad (1.33)$$

where N is the number of colors and d is the dimension of the spacetime before compactification. This result immediately follows from the invariance of the path integral under the translation $X_\mu \longrightarrow X_\mu + \epsilon X_\mu$.

2 Markov Chain Monte Carlo

2.1 Importance sampling

In any Monte Carlo integration the numerical error is proportional to the standard deviation of the integrand and is inversely proportional to the number of samples. Thus in order to reduce the error we should reduce the standard deviation as it does not require any extra computation. This is done by sampling the important regions of the integral where the integrand is largest.

Lets consider the one dimensional integral

$$I = \int_a^b f(x) dx. \quad (2.1)$$

We also introduce the probability distribution $p(x)$ such that

$$\int_a^b p(x) dx = 1. \quad (2.2)$$

The integral can be written as

$$I = \int_a^b p(x) \frac{f(x)}{p(x)} dx. \quad (2.3)$$

The above integral can be evaluated by sampling the random number according to the probability distribution $p(x)$. If we have set of N random numbers x_i which are distributed according to the $p(x)$, then the above integral can approximately written as

$$I_N = \frac{1}{N} \sum_{i=1}^N \frac{f(x_i)}{p(x_i)}. \quad (2.4)$$

The probability distribution $p(x)$ is chosen such that $\frac{f(x)}{p(x)}$ is slowly varying as this will reduce the standard deviation.

2.2 Metropolis algorithm

Suppose we are given the partition function

$$Z = \int [\mathcal{D}\phi] e^{-S[\phi]}, \quad (2.5)$$

and we want to evaluate average value of some observable $\mathcal{O}(\phi)$ given by

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int [\mathcal{D}\phi] \mathcal{O}(\phi) e^{-S[\phi]}. \quad (2.6)$$

Since this is a infinite dimensional integral and it can be evaluated analytically for special cases only, so we need to evaluate this numerically. One way is to discretize the integral and this integral becomes sum over the microstates (here field configurations) of the above quantity, i. e.

$$\langle \mathcal{O} \rangle \simeq \langle \mathcal{O} \rangle_n = \frac{\sum_{s=1}^n \mathcal{O}(\phi_s) e^{-S[\phi_s]}}{\sum_{s=1}^n e^{-S[\phi_s]}}. \quad (2.7)$$

In our case the number of microstates are infinite and calculating this with finite n will give undesirable result if we start calculating it without any proper algorithm. The reason is we do not know which microstates contribute the most to the integral.

So we need to use importance sampling to get the field configurations which contribute largely to the integral. Now we need to introduce a probability distribution p_s and we can rewrite the average value $\langle \mathcal{O} \rangle$ as

$$\langle \mathcal{O} \rangle = \frac{\sum_s \frac{\mathcal{O}(\phi_s)}{p_s} e^{-S[\phi_s]} p_s}{\sum_s \frac{e^{-S[\phi_s]}}{p_s} p_s}. \quad (2.8)$$

Now we generate the microstates s with the probability p_s and approximate $\langle \mathcal{O} \rangle$ with $\langle \mathcal{O} \rangle_n$ as

$$\langle \mathcal{O} \rangle_n = \frac{\sum_{s=1}^n \frac{\mathcal{O}(\phi_s)}{p_s} e^{-S[\phi_s]}}{\sum_{s=1}^n \frac{e^{-S[\phi_s]}}{p_s}}. \quad (2.9)$$

The *Metropolis algorithm* is importance sampling with p_s given by the Boltzmann distribution, i. e.

$$p_s = \frac{e^{-S[\phi_s]}}{\sum_{s=1}^n e^{-S[\phi_s]}}. \quad (2.10)$$

Then the above average value $\langle \mathcal{O} \rangle_n$ just becomes the arithmetic average

$$\langle \mathcal{O} \rangle_n = \frac{1}{n} \sum_{s=1}^n \mathcal{O}(\phi_s). \quad (2.11)$$

The Metropolis algorithm for our system is as follows.

- 1) Choose the initial field configuration. Random field configuration is known as hot start and if all the field configuration is set to 0, then it is known as cold start.
- 2) Then choose any random direction in field configuration space and make a jump in field configuration and lets call this ϕ_{trial} .
- 3) Then compute $\Delta S = S[\phi_{\text{trial}}] - S[\phi_{\text{old}}]$.
- 4) If $\Delta S \leq 0$, then accept this field configuration.
- 5) If $\Delta S > 0$, then choose a random number r in the interval $[0, 1]$ and if $r < e^{-\Delta S}$ then accept the field configuration, otherwise reject it.
- 6) Repeat step 2) to 5) for many times. This repetition of the steps depends on the how accurate we want our results to be. Thermalization should lie between these steps.
- 7) Calculate the average of the observable from the field configuration after the thermalization process.

2.3 Statistical errors

In any ergodic process we obtain a sequence of field configuration $\phi_1, \phi_2 \dots \phi_n$. We assume that all these configurations are thermalized configurations. Then the average value of the observable f and the statistical error δf are given by

$$\langle f \rangle = \frac{1}{n} \sum_{i=1}^n f_i, \quad (2.12)$$

$$\delta f = \frac{\sigma}{\sqrt{n}}, \quad (2.13)$$

where σ is the standard deviation. It is assumed that all the f_i are not correlated. This can be achieved by calculating the auto-correlated time τ_{int} and taking the field configurations after τ_{int} steps.

3 Simulation results

We calculated two observables for the pure Yang-Mills action, $\langle S \rangle$ and $\langle \text{Tr}(X_i^2) \rangle$. In the simulation, generators of the group are in anti-hermitian representation i. e., $\text{Tr}(T_a T_b) = -\delta_{ab}$. For the simulation to be fast and for the result to be considerably accurate the jump parameter δ is set such that acceptance rate is between 60% to 80%.

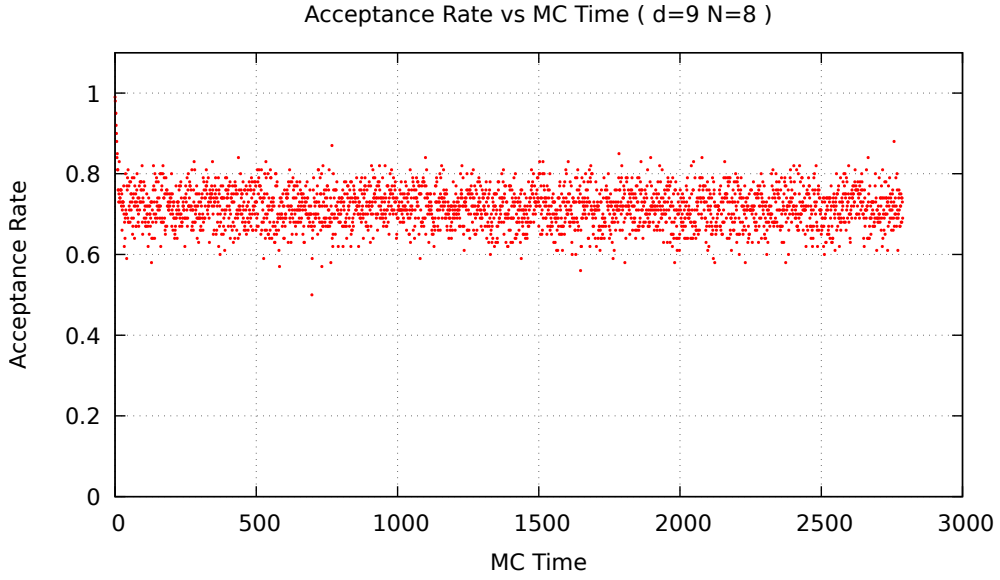


Figure 3.1: Acceptance rate against Monte Carlo time.

Fig. 3.1 shows the plot of acceptance rate against Monte Carlo time for $d = 9$ and $N = 8$ where both d and N has the meaning with respect to Eq. 1.33.

3.1 Average action $\langle S \rangle$

Value of $\langle S \rangle$ is calculated for different d and N . Fig. 3.2 shows the thermalization of the action for different d and N values.

In Fig. 3.3, we show the plot of $\frac{\langle S \rangle}{N^2 - 1}$ against d for different values of N .

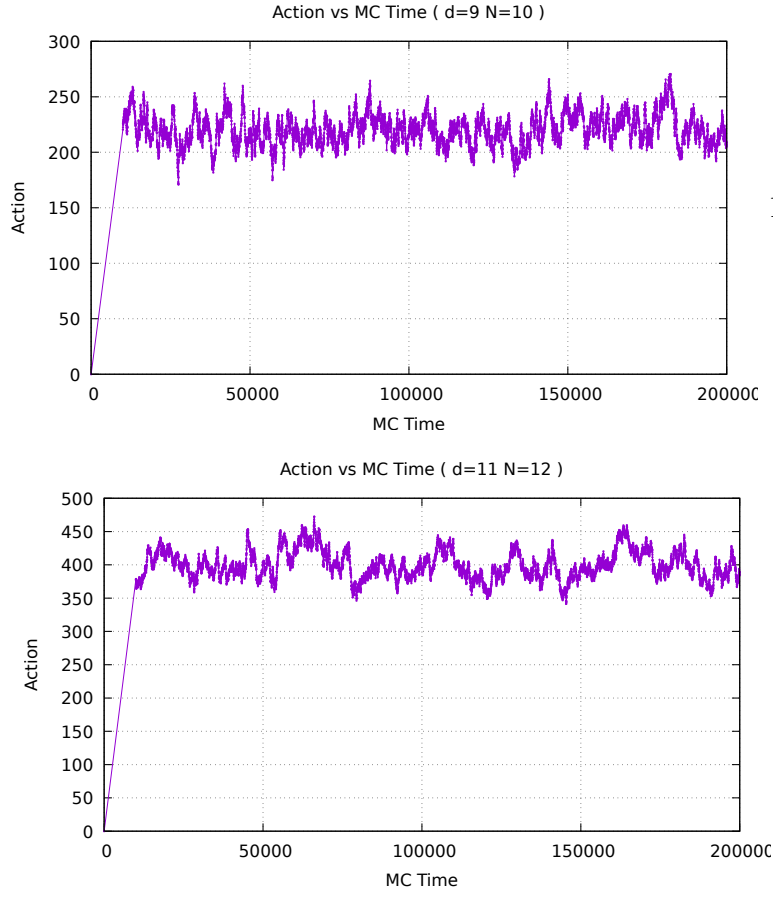
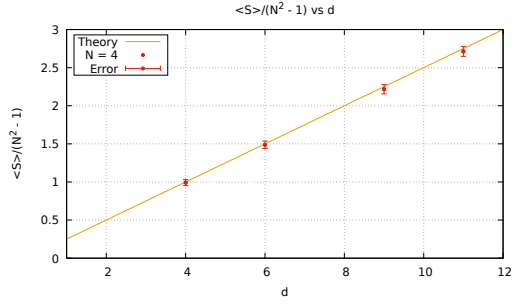


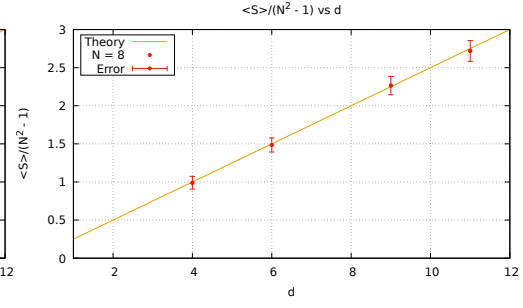
Figure 3.2: Plots showing thermalization of S with Monte Carlo time

In Fig. 3.4, we show the combined plot of $\frac{\langle S \rangle}{N^2 - 1}$ against d for different values of N .

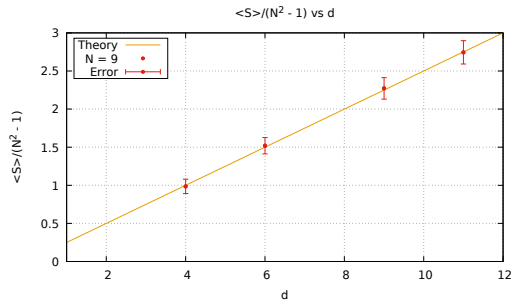
From Fig. 3.3 and Fig. 3.4 we can clearly see that the simulation result is in excellent agreement with analytical result.



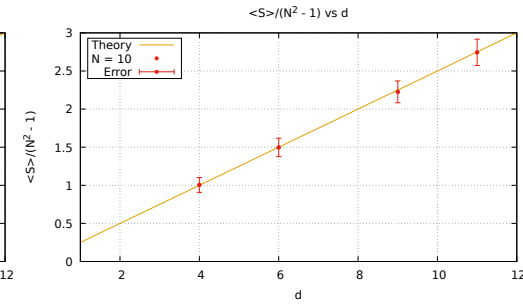
(a) $N=4$



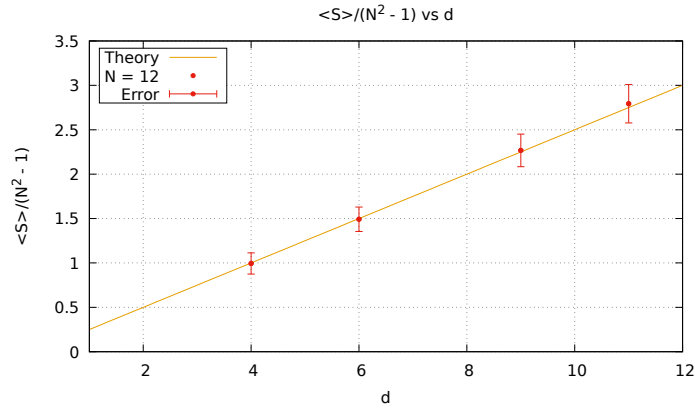
(b) $N=8$



(c) $N=9$



(d) $N=10$



(e) $N=12$

Figure 3.3: Plot of $\frac{\langle S \rangle}{N^2 - 1}$ against d for different values of N .

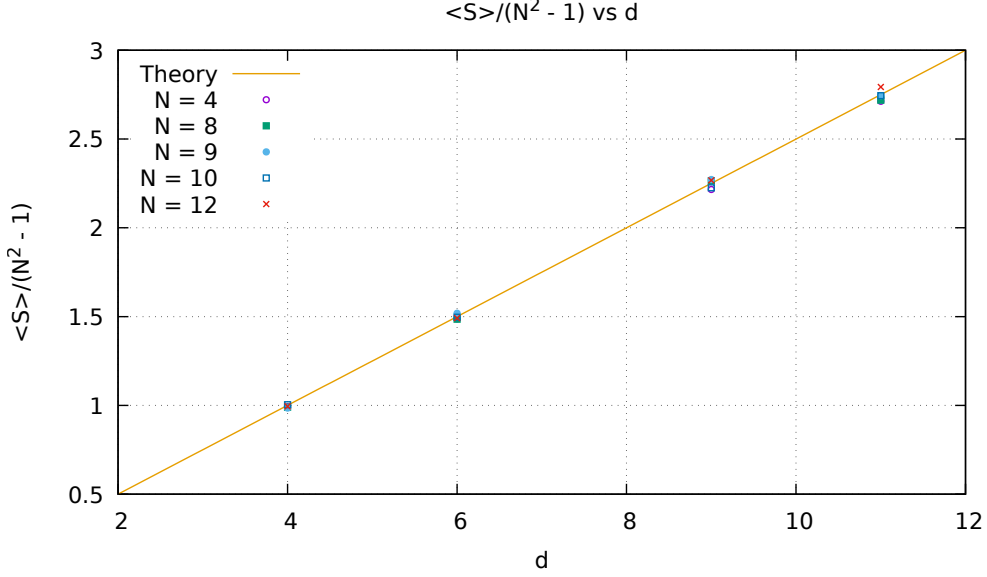


Figure 3.4: Combined plot of $\frac{\langle S \rangle}{N^2 - 1}$ against d for different values of N .

3.2 Extent of spacetime R

We can define another observable in the model, called the extent of spacetime

$$R = \sqrt{\frac{\langle \text{Tr}(X_i^2) \rangle}{N}}. \quad (3.1)$$

The value of R^2 is calculated for different d and N . We plot the thermalization of $\text{Tr}(X_i^2)$ for different d and N in Fig. 3.5. In Fig. 3.6, we show the plot of R^2 against d for different values of N . The combined plot of R^2 against d for different N is shown in Fig. 3.7.

In Ref. [3] the authors provide the behavior of the extent of spacetime for large N as

$$R \sim N^\omega, \quad (3.2)$$

where an upper bound of ω for bosonic matrix model was found to be

$$\omega \leq \frac{1}{4} \quad (3.3)$$

by perturbative arguments.

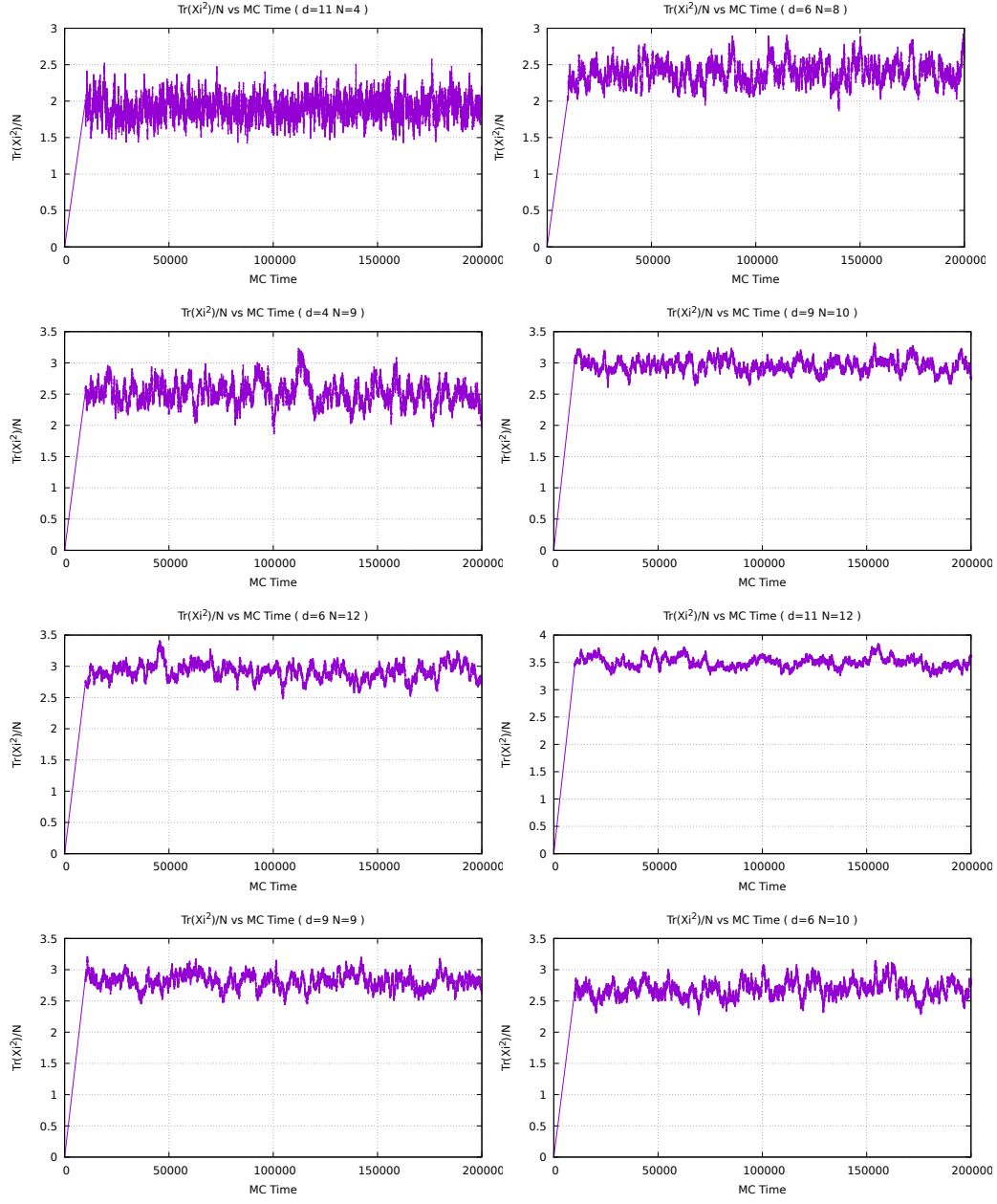
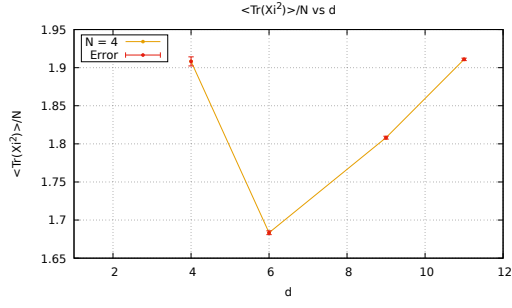
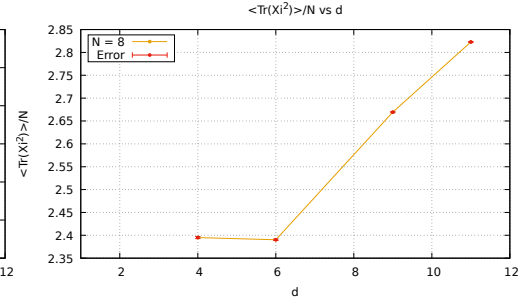


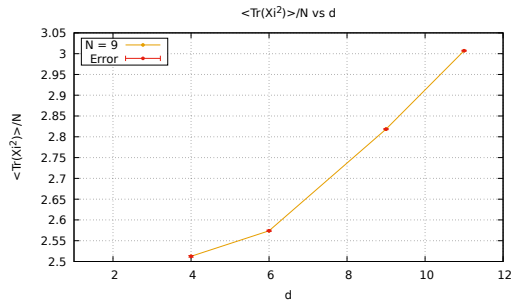
Figure 3.5: Plots showing thermalization of $\text{Tr}(X_i^2)$ with Monte Carlo time.



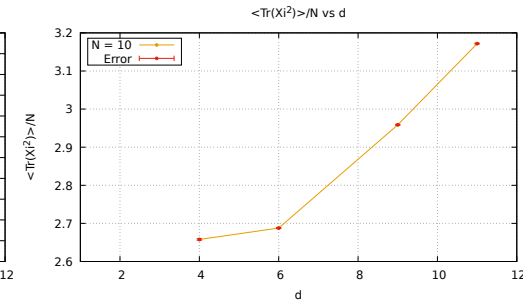
(a) N=4



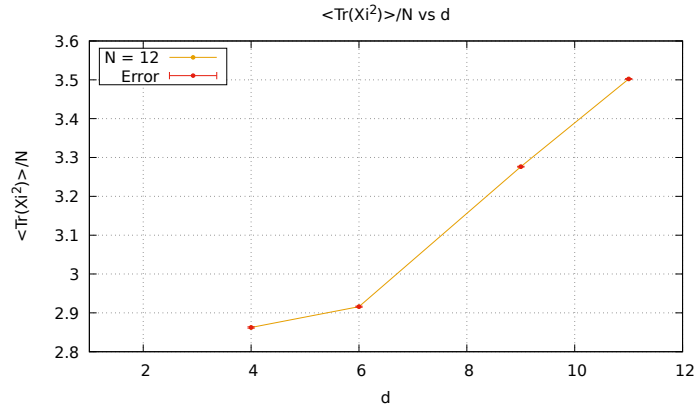
(b) N=8



(c) N=9



(d) N=10



(e) N=12

Figure 3.6: Plot of $\frac{\langle \text{Tr}(X_i^2) \rangle}{N}$ against d for different values of N .

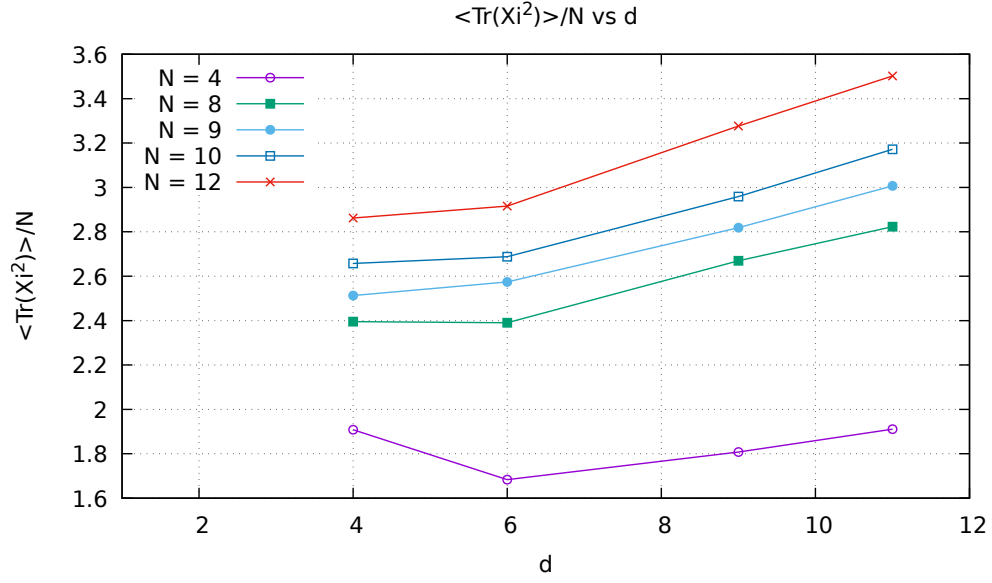


Figure 3.7: Combined plot of $\frac{\langle \text{Tr}(X_i^2) \rangle}{N}$ against d for different values of N .

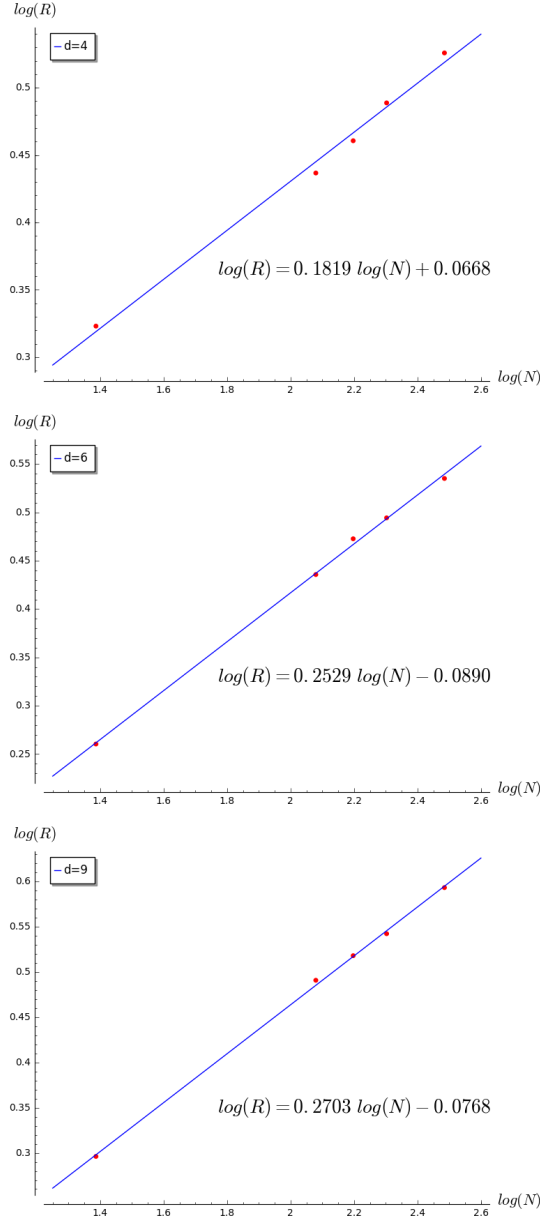


Figure 3.8: The extent of spacetime R against N for various values of d . The solid lines show linear fits to the data. We see that the slope $\omega \lesssim \frac{1}{4}$, as predicted by the behavior in Ref. [3]

d	ω
4	0.1819(293)
6	0.2529(168)
9	0.2703(152)

Table 1: The values of ω from linear fit of data for various d . The analytical prediction is $\omega \leq \frac{1}{4}$.

4 Discussions and conclusions

In this project we have simulated zero-dimensional $SU(N)$ Yang-Mills matrix model with number of colors $N = 4, 8, 9, 10, 12$ and number of scalars ranging from $d = 4, 6, 9, 11$. We have computed the expectation value of the Euclidean action, $\langle S \rangle$ and the extent of spacetime R . These two observables are in excellent agreement with the analytical results. In the simulations, we have not included the Chern-Simons term as it is a purely imaginary term, and Metropolis algorithm cannot handle imaginary parts of the action. In the future we can simulate the above system with the Chern-Simons term using a different algorithm, for example Complex Langevin with stochastic quantization.

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